



INDIAN INSTITUTE OF TECHNOLOGY KANPUR

CGS402: Applied Cognitive Science

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Under the Guidance of

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Submitted by - Group A

Ed-Tech App Survey on Cognitive Principles:

coursera

Task Allocation

Member	Task
Manan Patel (220608)	[Memory]: Encoding, Engagement & Initial Learning
Aditya Prakhar (230076)	[Memory]: Feedback, Retention & Metacognitive Awareness
Vanshika Yadav (221168)	[Neilson's Principles]: Aesthetic and Minimalist Design, Help Users Recognize, Diagnose, and Recover from Errors & Help and Documentation
Sapna (220973)	[Neilson's Principles]: Consistency and Standards, Error Prevention, Recognition Rather than Recall & Flexibility and Efficiency of Use
Garima (220396)	[Neilson's Principles]: Visibility of System Status, Match Between System and the Real World & User Control and Freedom
Brinda Fadadu (230307)	[Real world problems]: Engagement & Motivation, Learning Effectiveness
Suneet Maharana (221105)	[Real world problems]: Access & Equity, Social & Competitive Dynamics
Pragati Agrawal (220779)	[Real world problems]: Trust & Long-Term Use, Net Promoter Score

Introduction

Overview of the EdTech Application

Coursera is a leading online learning platform that offers courses, professional certificates, and degree programs from top universities and organizations worldwide. Founded in 2012, it provides learners with flexible, self-paced education across a wide range of subjects, including science & technology, business, and the humanities. Coursera combines video lectures, interactive assignments, and peer-reviewed assessments to create an engaging learning experience, making quality education accessible to millions of users globally.

Rationale for Selecting Coursera

We selected Coursera for this analysis because of its widespread use among students on campus and its extensive range of courses spanning multiple disciplines. A key reason for this choice is how the platform integrates several learning principles studied in cognitive science—such as structured content delivery, active recall through quizzes, and progressive skill building—into its course design. Additionally, its global reach and collaborations with leading universities and organizations position it as a major player in online education, making it a strong and relevant candidate for in-depth study and analysis.

Details of Data Collected

Our study cohort consisted of 28 participants, representing a demographic of young adults ranging from 16 to 23 years old. The gender distribution within the sample was relatively balanced, with 16 male (57.1%) and 12 female (42.9%) respondents. Academically, the group exhibited a diverse level of background knowledge, reporting a moderate mean familiarity with Cognitive Science of 2.86 on a 5-point scale.

Survey Resources

[Survey Form](#) | [Collected Data](#)

Memory

Q1: How Is New Material Presented?

[Dual-Encoding]

Question: “When you first encounter new material in the app, how is it presented?”

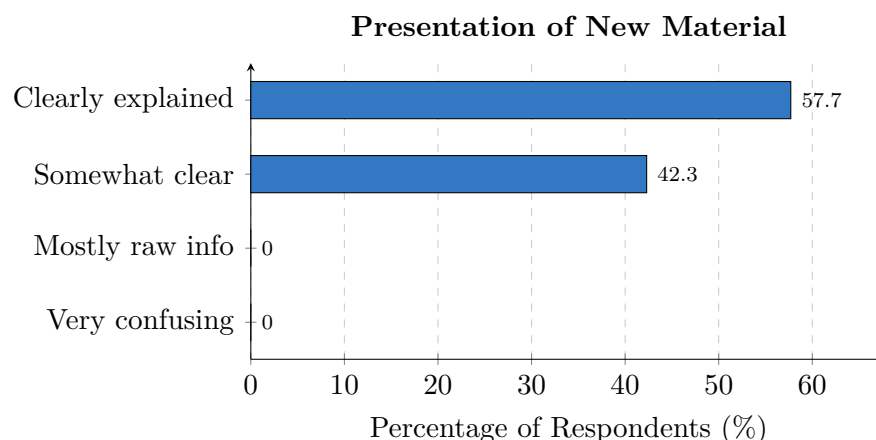


Figure 1: Q1 – How is new material presented? Categorical responses ($n = 26$).

Results: A majority (57.7%) found new material clearly explained with examples, visuals, and conceptual connections. The remaining 42.3% found the explanation only partial. No respondent reported raw or confusing presentation.

Semantic Processing and Dual-Encoding:

The nearly equal split between “clearly explained” (57.7%) and “somewhat clear” (42.3%) reflects variation in encoding quality across courses. Strong explanations support semantic processing (Craik and Lockhart, 1972), where deeper understanding leads to better retention than surface-level learning. Coursera’s use of both verbal explanations and visuals aligns with Paivio’s Dual-Encoding Hypothesis, which enhances recall by creating multiple memory traces. However, the “somewhat clear” responses indicate that this level of encoding is not consistent. Standardizing content ensuring each concept includes examples and visual aids can improve overall clarity

Q2: Gamification and Focus

[Processing Efficiency Theory]

Question: “How do game elements (points, levels, rewards) affect your focus?”

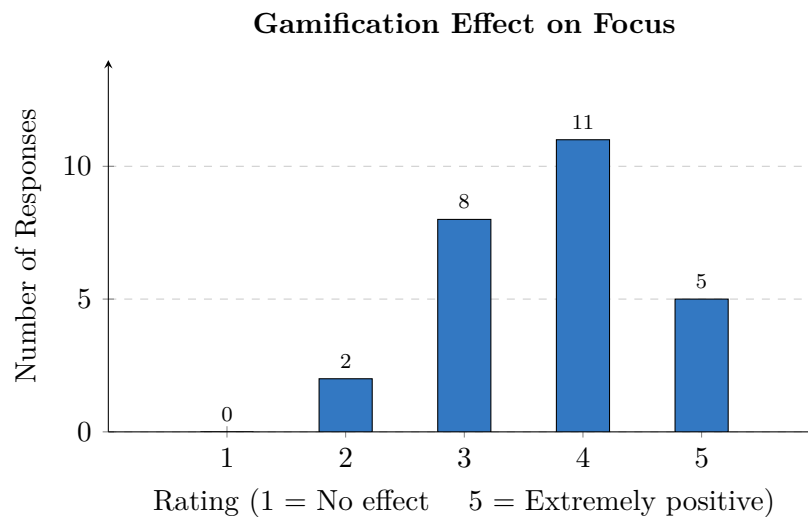


Figure 2: Q2 – Effect of gamification on focus, Likert 1–5 ($n = 26$). **Mean = 3.73.**

Results: No respondent rated game elements as completely ineffective. Ratings cluster at 4 (42.3%) and 3 (30.8%), with 19.2% at the maximum. Mean = **3.73 / 5**.

Processing Efficiency Theory and Motivational Salience:

Game elements like points, streaks, and badges act as extrinsic motivators that help maintain attention and engagement. This is reflected in the relatively high mean score of 3.73, indicating that they are generally effective in maintaining focus. According to Processing Efficiency Theory (Eysenck & Calvo, 1992), such motivational cues can enhance focus when kept at optimal levels, as these game elements help ensure that attention remains directed toward the task. However, for intrinsically motivated users, excessive reliance on these rewards may reduce internal motivation, consistent with the Over-Justification Effect (Deci & Ryan).

Q3: Does the App Revisit Previously Learned Material?

[Spacing Effect]

Question: “Does the app revisit previously learned material after days/weeks?”

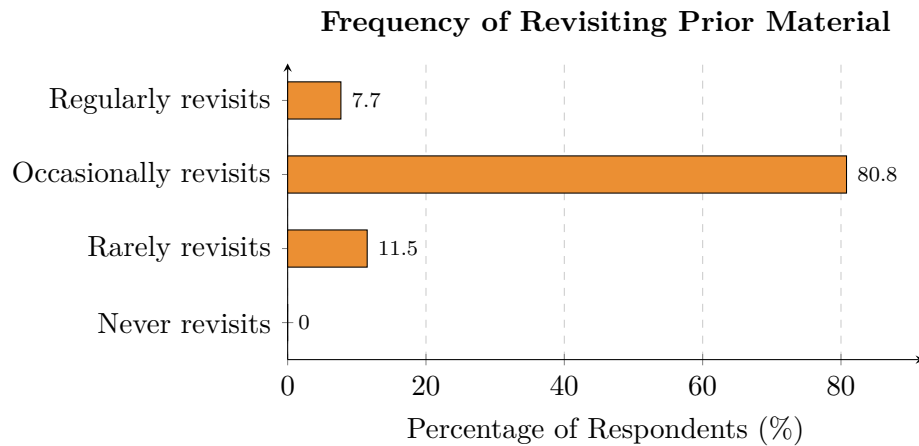


Figure 3: Q3 – Frequency of spaced content revisitation ($n = 26$).

Results: The dominant response is *Occasionally revisits* (80.8%). Only 7.7% experience regular revisitation and 11.5% report it rarely occurs. No respondent selected Never.

Spacing Effect and the Forgetting Curve:

Ebbinghaus’ forgetting curve shows that memory fades quickly without revision, while spaced review helps retain information longer. The Spacing Effect suggests that reviewing material at intervals (like after one day, one week, and one month) is far more effective than cramming. The 80.8% “occasionally” response indicates that Coursera does not consistently use spaced repetition, and any revision happens unintentionally. Introducing a simple spaced review system with reminders at fixed intervals could significantly improve retention with minimal effort.

Q4: How Meaningful Is the Content Review?

[Elaborative Retrieval]

Question: “When content is repeated, how meaningful is the review?”

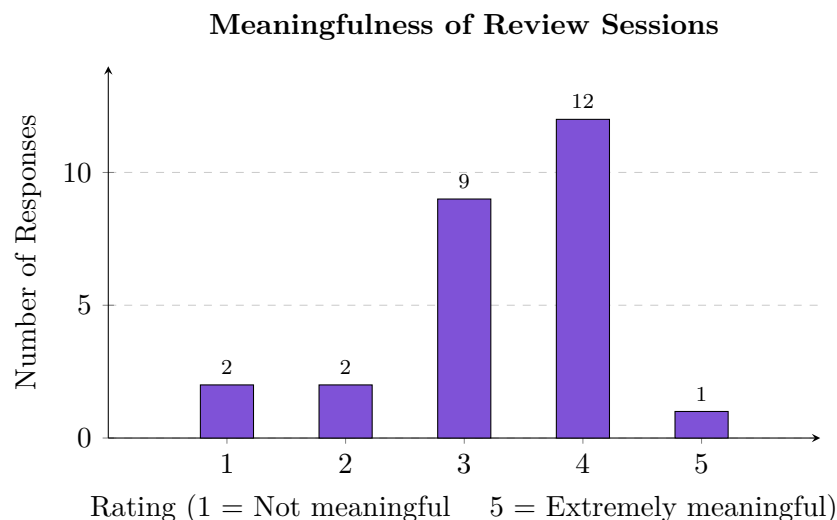


Figure 4: Q4 – Perceived meaningfulness of review, Likert 1–5 ($n = 26$). **Mean = 3.31.**

Results: The mode falls at rating 4 (46.2%), but notable responses at 1 and 2 (each 7.7%) pull the mean to **3.31 / 5**. Only 3.8% rated review as extremely meaningful.

Passive Re-exposure vs. Elaborative Retrieval:

Revisiting should ideally have a high impact on memory, as active recall is one of the most effective ways to strengthen retention. However, not all forms of revisiting are equally useful. Simply rereading or rewatching lectures has limited effect compared to active, elaborative retrieval (Roediger and Karpicke, 2006). The relatively low average score of 3.31 suggests that Coursera’s revisiting methods rely more on passive approaches like summaries or progress reviews rather than tasks that require active recall. Incorporating generative quizzes for example, asking learners to summarize key ideas from memory can significantly improve retention.

Q5: How Does the App Test Understanding?

[Testing Effect]

Question: “How does the app test your understanding?”

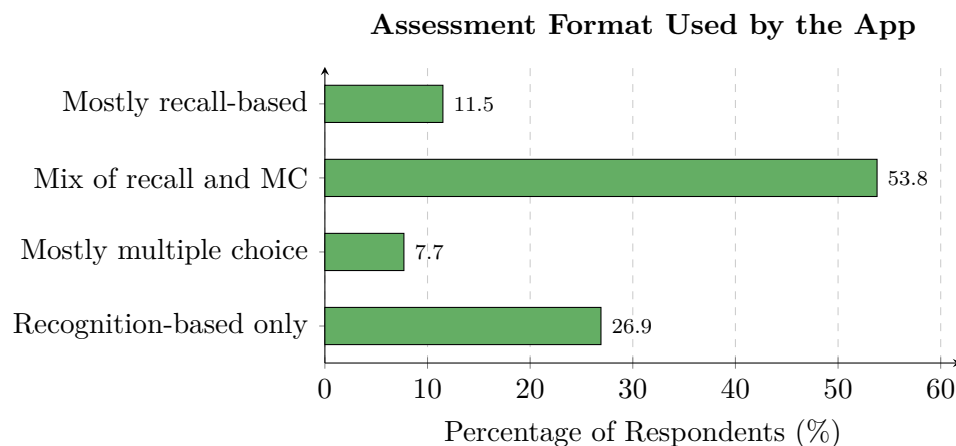


Figure 5: Q5 – Perceived assessment format distribution ($n = 26$).

Results: Over half (53.8%) perceived a balanced mix of recall and multiple-choice formats. Notably, 26.9% characterised assessments as purely recognition-based, while only 11.5% reported primarily free-recall formats.

The Testing Effect and Desirable Difficulties:

Free-recall testing has been shown to result in significantly better retention than recognition tests by Roediger and Karpicke (2006). This is due to the Testing Effect, in which retrieving information makes it easier to retrieve in the future. The percentage of participants who view assessment questions as recognition tests, at 26.9%, is troubling, since such types of assessment questions remove the retrieval cue and thus reduce the difficulty of the test and minimize the testing effect.

Q6: What Happens After a Challenge?

[Transfer / Encoding Specificity]

Question: “After completing a challenge, what happens next?”

Results: The largest group (57.7%) reports the app simply presents the correct answer and advances. About a third (30.8%) receive a varied follow-up question, and only 11.5% are asked to apply the concept in a novel context.

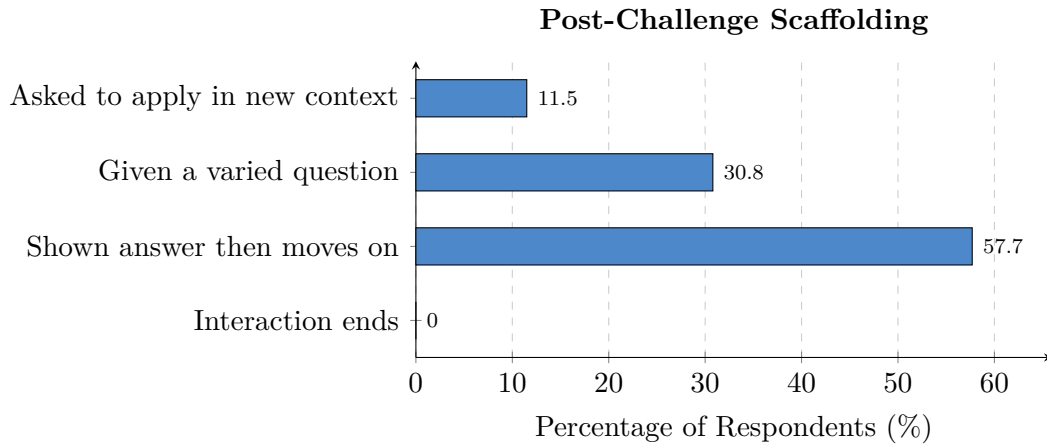


Figure 6: Q6 – What happens after completing a challenge ($n = 26$).

Transfer-Appropriate Processing and Encoding Specificity:

The post-challenge stage is crucial for deep learning. Solving only one problem with the answer provided leads to minimal elaborative encoding and does not support transfer appropriate encoding. According to the Principle of Encoding Specificity (Tulving and Thomson), memory is best retrieved in the same context in which it was learned, whereas transfer requires applying knowledge in a new context. This aligns with the Encoding Variability Principle, where exposure to varied contexts strengthens abstraction and generalization. However, only 11.5% of learners are exposed to such effective post-challenge practice.

Q7: How Helpful Is the Error Explanation?

[Error-Driven Learning]

Question: “When you make a mistake, how helpful is the explanation?”

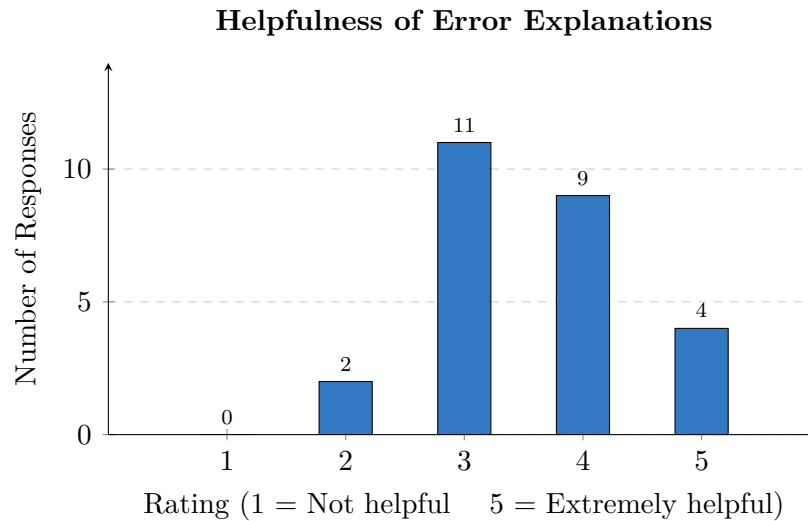


Figure 7: Q7 – Helpfulness of error explanations, Likert 1–5 ($n = 26$). **Mean = 3.58.**

Results: No respondent rated explanations as unhelpful. The distribution peaks at rating 3 (42.3%), with 34.6% at 4 and 15.4% at 5. Mean = **3.58** / 5.

Error-Driven Learning and Corrective Feedback:

Committing mistakes is a highly effective stimulus to memory encoding (Metcalf, 2017), but only when supplemented with *explanatory* feedback rather than mere answer validation. Mistakes trigger prediction error messages that prepare the brain for modification of the memory trace in question yet this modification is only possible with substantial information regarding the reasons for the incorrect answer. The skew toward a rating of 3 suggests that the feedback or explanations provided for incorrect answers on Coursera are not very effective, and learners often do not fully understand their mistakes from them. Improving the quality of feedback by making it more explanatory and concept focused would help learners better grasp the underlying principles.

Q8: How Effective Is the Timing of Feedback?

[Feedback Timing]

Question: “How effective is the timing of feedback?”

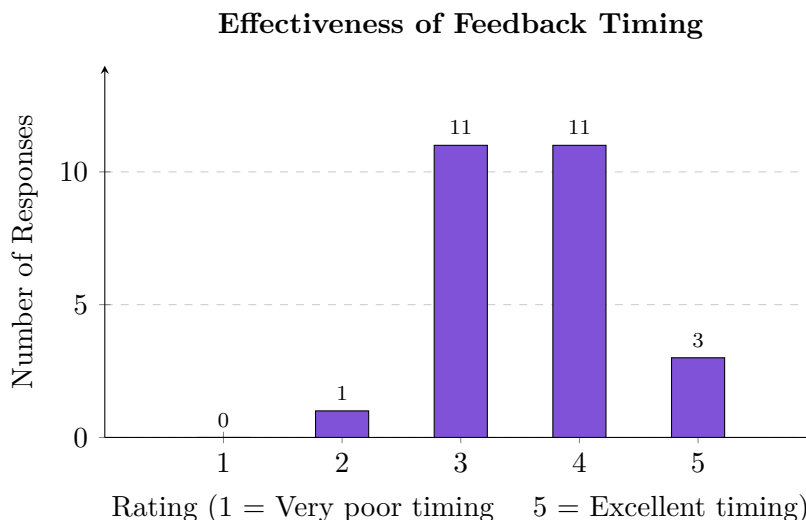


Figure 8: Q8 – Feedback timing effectiveness, Likert 1–5 ($n = 26$). **Mean = 3.62.**

Results: Ratings of 3 and 4 are tied at 42.3% each, with 11.5% at 5. Only one respondent rated timing at 2; none at 1. Mean = **3.62 / 5**.

Immediate vs. Delayed Feedback in Memory Consolidation:

The ratings, with 42.3% at both 3 and 4 and a mean of 3.62, suggest that feedback timing on Coursera is generally effective but not consistently optimal. According to cognitive science, immediate feedback is more effective for error correction in low-stakes tasks (Kulik and Kulik, 1988), as the mental context is still fresh and the feedback can be easily linked to the task. This aligns with the relatively higher ratings. However, the absence of very high ratings and the presence of mid-level responses indicate that delayed feedback such as in peer reviews is less effective, since the original mental context may fade (Butler and Roediger, 2008). Providing more immediate feedback, especially in quizzes, could improve overall effectiveness.

Q9: Confidence in Long-Term Retention

[Decay with Disuse]

Question: “How confident are you that you would remember the content after 2 weeks (outside the app)?”

Results: This item records the **lowest mean of all ten questions: 2.96 / 5**. No respondent gave a maximum rating. The distribution: 7.7% rated 1, 23.1% rated 2, 34.6% at 3, and 34.6% at 4.

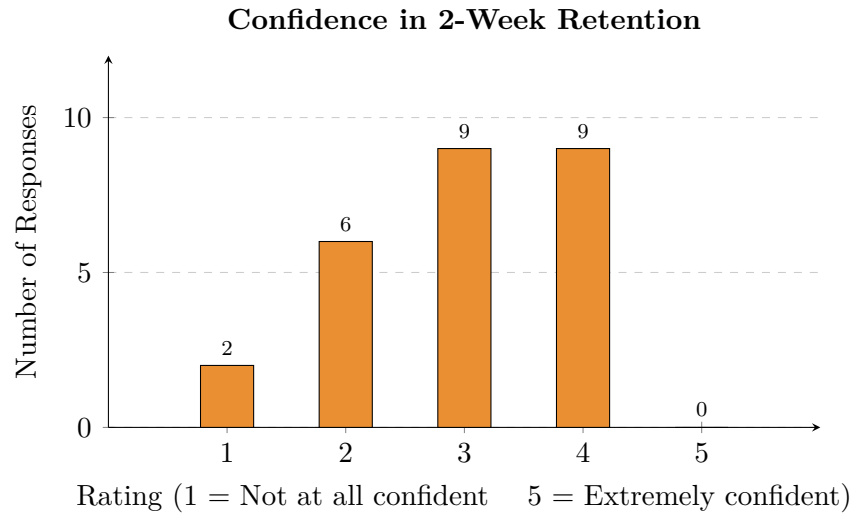


Figure 9: Q9 – Self-rated confidence in retaining content after two weeks ($n = 26$). **Mean = 2.96.**

Decay with Disuse – The Most Critical Finding:

The low mean of 2.96, along with no responses at the maximum rating, indicates that learners lack confidence in long-term retention. This can be explained by decay due to disuse, where information fades without repeated retrieval (Ebbinghaus, 1885). The distribution, with most responses clustered at 3 and 4 but none at 5, suggests partial understanding but weak consolidation. Coursera’s model does not consistently reinforce learning through spaced retrieval or active practice, leading to rapid forgetting. The absence of systematic review mechanisms (e.g., spaced repetition or application-based tasks) prevents strengthening of memory traces, resulting in low confidence after two weeks.

Q10: Is Learning Progress Visible?

[Metacognitive Monitoring]

Question: “Has the app shown you your learning progress over time (e.g., progress checks, challenges)?”

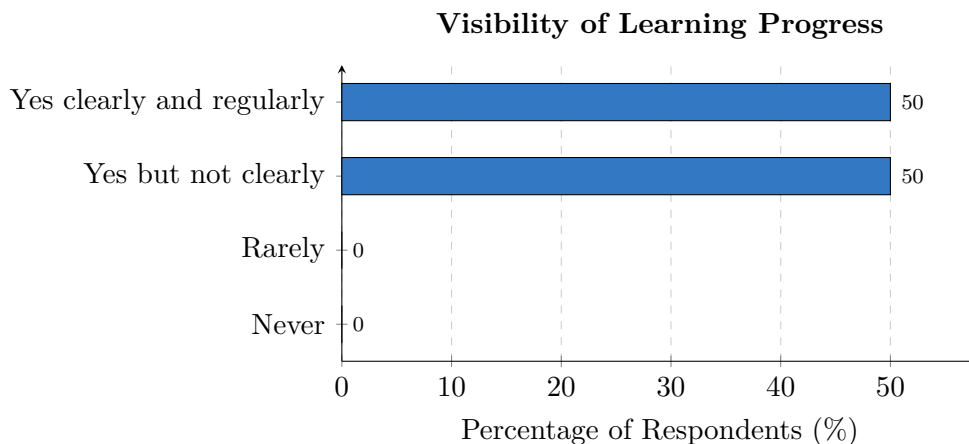


Figure 10: Q10 – How clearly is learning progress displayed ($n = 26$).

Results: Responses split exactly 50–50 between “Yes, clearly and regularly” and “Yes, but not very clearly.” No respondent selected Rarely or Never.

Metacognitive Monitoring and Visibility of System Status:

The visibility of progress serves two purposes: motivational support derived from seeing progress and metacognitive calibration – accurate awareness among learners about what they know and what they have yet to learn (Flavell, 1979). Metacognitive accuracy is a strong predictor of self-regulation and academic achievement. The 50% “not very clearly” indicates that, although there is a visibility of progress made in Coursera courses, the progress indicator is not detailed enough for metacognitive calibration. For instance, a percentage bar showing module completion provides information on the progress of watching a particular lecture, but offers no clue as to the extent of comprehension by the learner.

Nielson’s principles

Visibility of System Status

Q1. When you complete an activity or earn a reward, does the app make it immediately clear what just happened and what comes next or do you ever feel unsure whether your progress was saved?

Observations and Implications

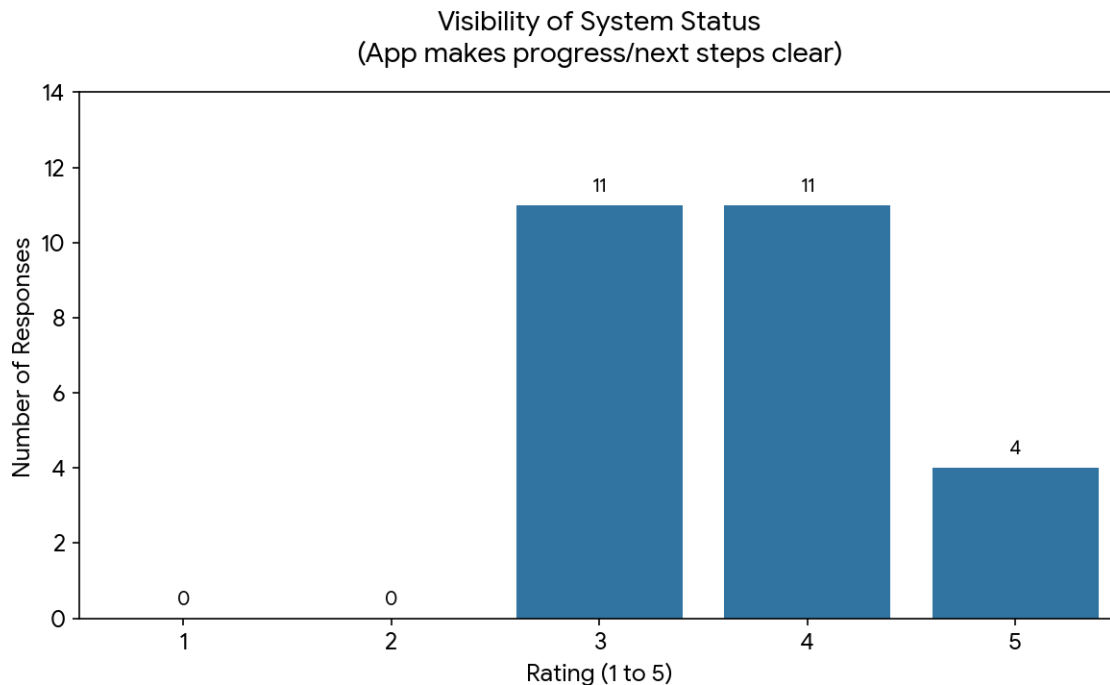


Figure 11: Are the next steps clear?

- The response distribution shows no negative ratings (1 or 2), with a majority of users (15/26) giving positive ratings (4 or 5) and the remaining 11 giving neutral ratings (3).
- The findings suggest that the app successfully avoids confusing users or making them feel entirely lost about their progress.
- However, the high volume of neutral scores implies that while the app meets baseline expectations, it lacks the explicit, immediate feedback required for complete user confidence.

Behavioural Analysis of the Users

The responses suggest the following behavioural patterns among users:

- Because feedback is not universally obvious, users might experience a micro-pause to verify their progress instead of transitioning seamlessly to the next task.
- Users do not experience severe anxiety over lost progress, but they may hesitate or double-check their status.
- The absence of clear visual rewards (like a “Progress Saved!” notification) means users miss out on psychological momentum, reducing the overall satisfaction of the interaction.

Match Between System and the Real World

Q2. Do the instructions, labels, and game concepts in the app feel natural and easy to understand, or do they use jargon that makes you pause or guess?

Observations and Implications

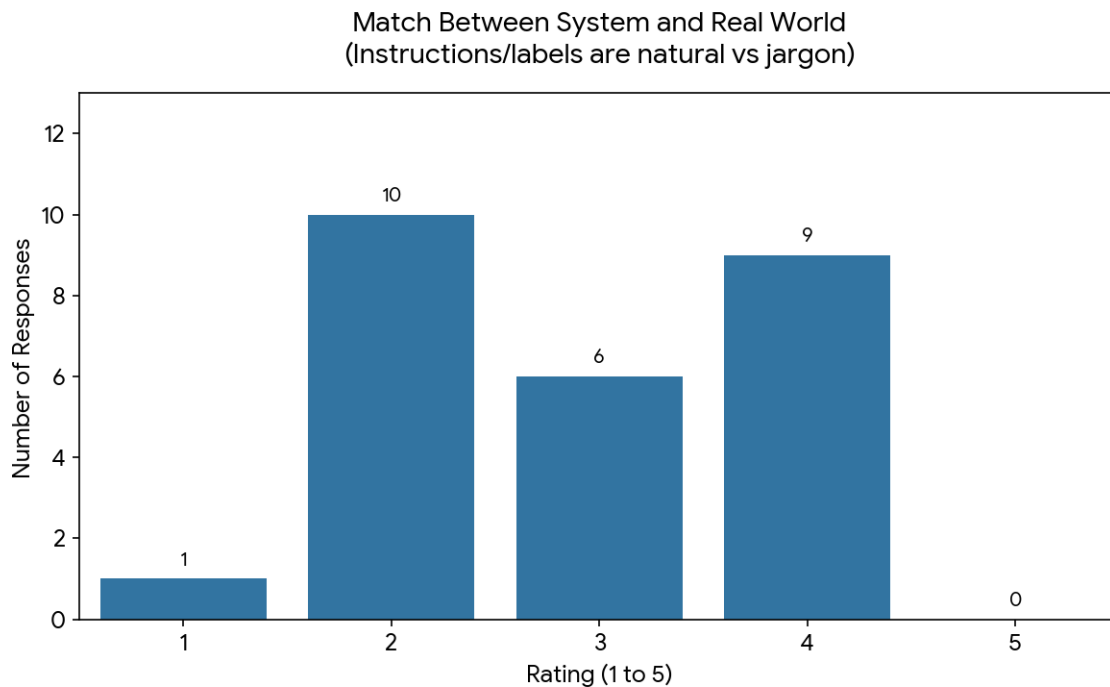


Figure 12: Does the Instructions/labels feels natural?

- The response distribution is highly polarized, with nearly 42% of participants (11/26) giving negative ratings, while 34% (9/26) gave positive ratings.
- The polarization indicates that the app’s terminology and gamified concepts act as a cognitive barrier for a significant portion of the user base.
- The findings imply that the app’s language aligns with the mental models of a specific demographic (likely those with prior gaming experience) while completely alienating others.

Behavioural Analysis of the Users

The responses suggest the following behavioural patterns among users:

- Users unfamiliar with the app’s specific jargon are forced to pause and guess meanings, which introduces unnecessary cognitive load and makes learning much harder.
- When instructions do not map to everyday, real-world concepts, users often resort to trial-and-error clicking just to see what happens.
- Repeated exposure to unnatural terminology can frustrate users over time, leading to fatigue and an increased likelihood of abandoning the app entirely.

User Control and Freedom

Q3. If you accidentally tapped the wrong answer or entered the wrong level, could you easily undo or go back or did you feel stuck with a mistake?

Observations and Implications

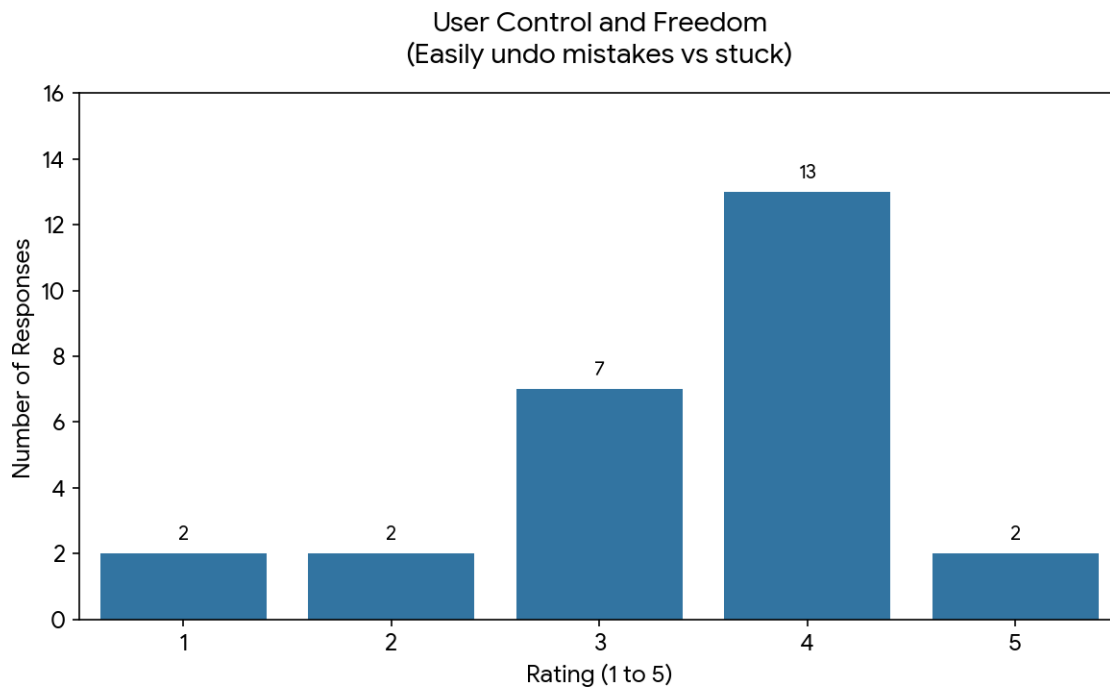


Figure 13: How easy it is to undo?

- A strong majority of users (15 out of 26) gave positive ratings, with only 4 users expressing frustration (rating 1 or 2).
- The results indicate that the app successfully provides intuitive “emergency exits” for the vast majority of its users.
- The findings imply that the overall architecture for returning to previous states or undoing actions is generally well-implemented and functional.

Behavioural Analysis of the Users

The responses suggest the following behavioural patterns among users:

- Knowing they can easily recover from minor mistakes or accidental taps, users interact with the app more confidently and swiftly.
- This sense of freedom encourages exploration within the app without the fear of permanent negative consequences or ruining their progress.
- However, the small subset of frustrated users suggests there may be isolated screens where the **back** or **undo** buttons are missing, causing brief moments of friction and making them feel trapped.

Consistency and Standards

Q4. Have you ever been confused because a button, icon, or interaction worked differently in one part of the app than in another?

Observations and Implications

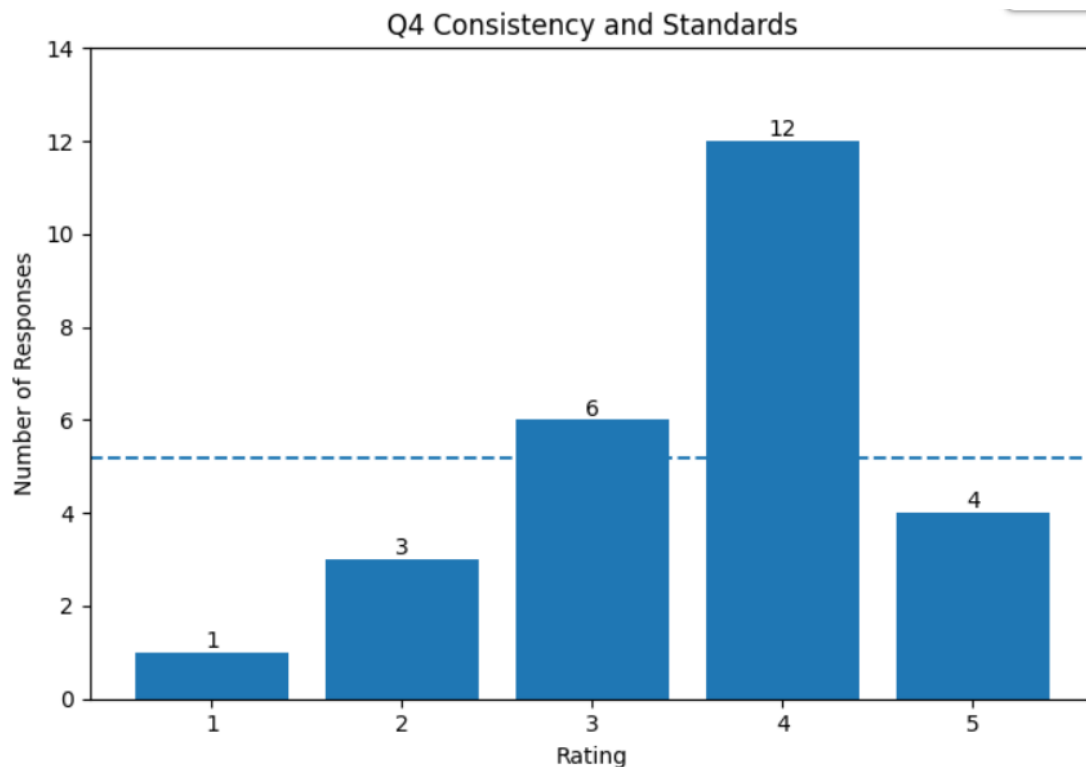


Figure 14: Participant responses for Q4

- The response distribution is skewed towards higher ratings, with the mean score at **3.58/5**, indicating noticeable confusion caused by inconsistent interface behaviour.
- Since this question is negatively framed, higher ratings imply poorer adherence to the principle of consistency and standards.
- The findings suggest that similar buttons, icons, or interaction patterns are not always interpreted as behaving the same way across different sections of the platform.

Behavioural Analysis of the Users

The responses suggest the following behavioural patterns among users:

- Users may need to relearn interface behaviour when moving across different sections, which increases cognitive effort and interrupts task flow.
- Inconsistent signifiers can weaken trust in the platform, as users become less certain about the consequences of their actions.
- Repeated exposure to inconsistency can make users hesitant, leading them to proceed more cautiously or engage in trial-and-error interactions.

Error Prevention

Q5. Does the app ever let you do something by accident that you didn't intend like submitting an incomplete answer or skipping a required step without warning you first?

Observations and Implications

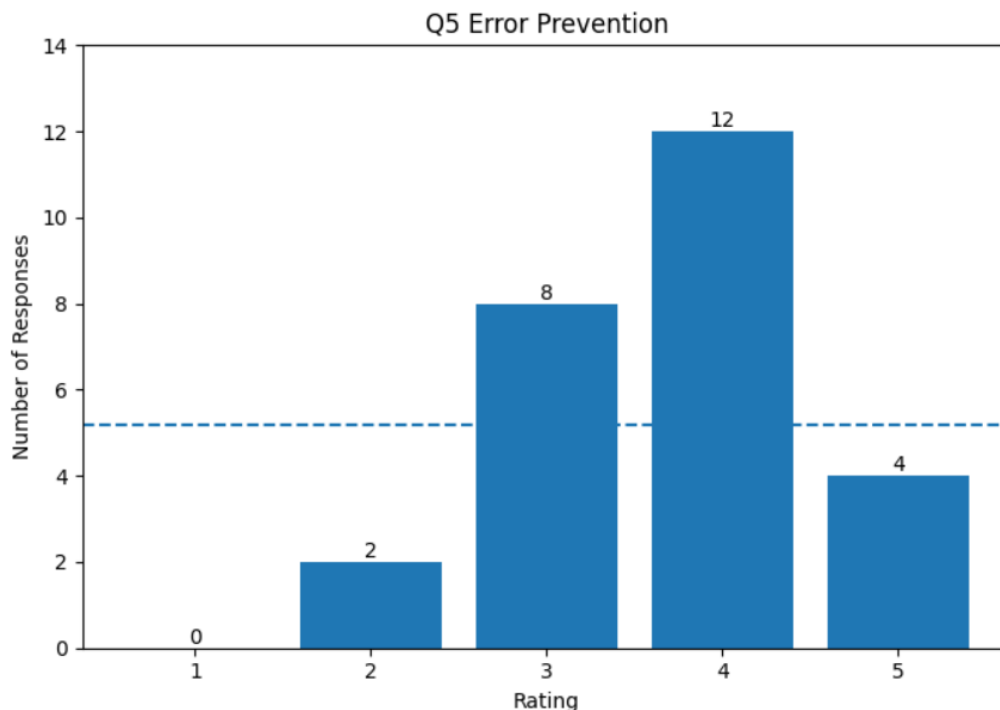


Figure 15: Participant responses for Q5

- The mean score for this item is **3.69/5**, showing that many users feel the platform allows unintended actions without adequate warning or confirmation.
- Because the question measures accidental actions, a higher score indicates weaker error prevention.
- The responses suggest that the system does not always stop users before they make avoidable mistakes such as incomplete submissions or unintended skips.

Behavioural Analysis of the Users

The observed pattern points to the following user behaviours:

- Users may rely on self-checking rather than the platform’s safeguards, which increases effort during interaction.
- Accidental actions can create frustration and reduce confidence in the correctness of completed tasks.
- Some users may become overly cautious after experiencing such errors, slowing down their interaction with the system.

Recognition Rather than Recall

Q6. When you return to the app after a break, can you quickly pick up where you left off, or do you have to remember a lot of context on your own?

Observations and Implications

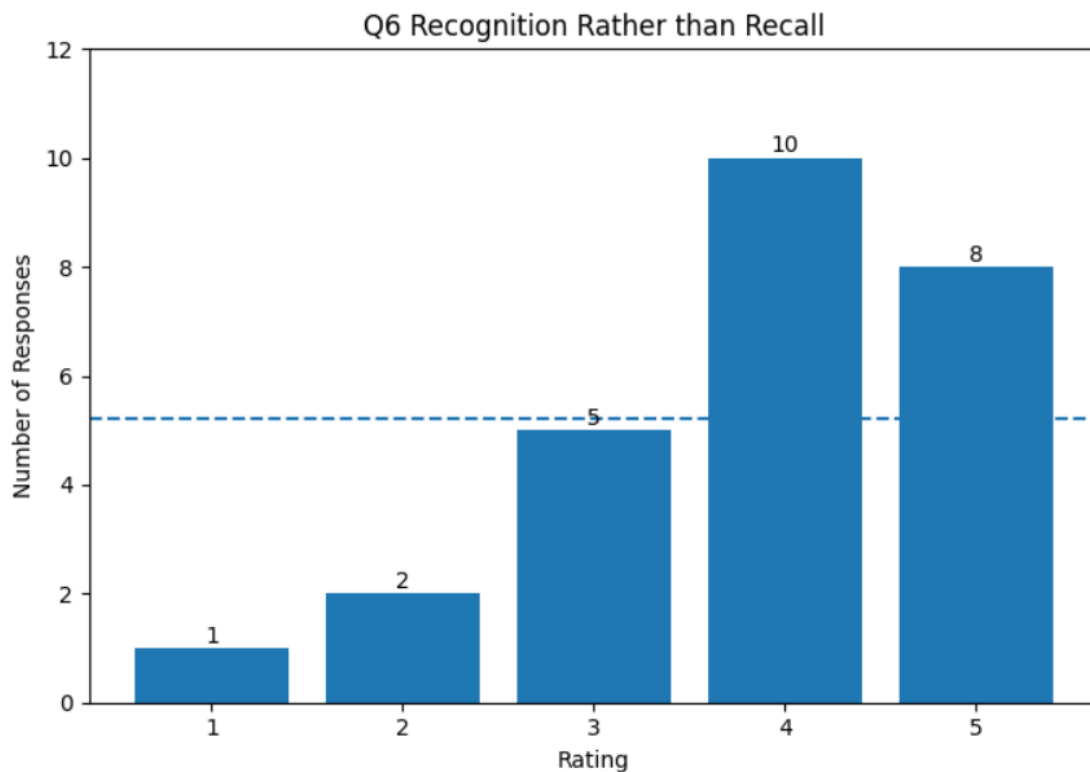


Figure 16: Participant responses for Q6

- This principle received the strongest performance among Q4–Q7, with a mean score of **3.85/5**.
- Higher ratings here represent better usability, implying that users are generally able to resume learning without having to remember too much prior context.
- The result suggests that visible cues such as progress indicators, structured modules, and navigation support are functioning reasonably well.

Behavioural Analysis of the Users

The response pattern indicates the following:

- Users are able to re-enter the learning flow with relatively low memory burden, reducing cognitive friction after interruptions or breaks.

- Visible structure likely supports confidence and continuity, making learners feel oriented within the platform.
- This kind of design can improve persistence by lowering the mental effort required to restart a session.

Flexibility and Efficiency of Use

Q7. Are there shortcuts, customization options, or ways to skip content you already know or does the app force everyone through the same path regardless of skill level?

Observations and Implications

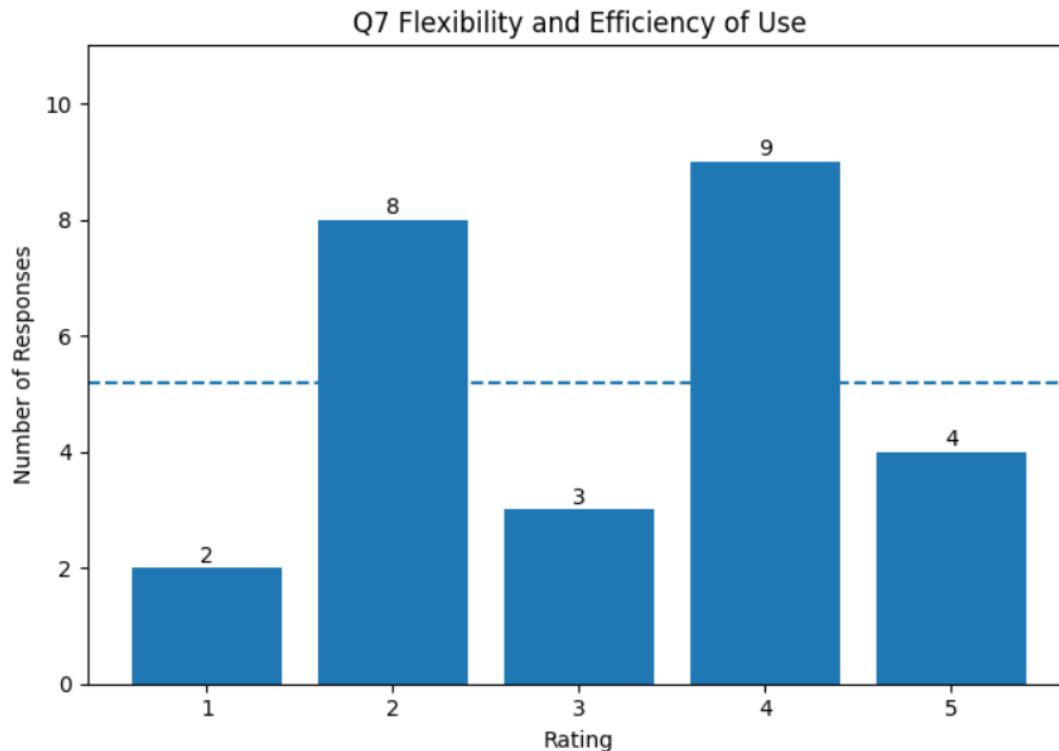


Figure 17: Participant responses for Q7

- The mean score of **3.19/5** indicates a mixed experience for users on flexibility and efficiency.
- Responses are spread across the scale, suggesting that while some users perceive useful flexibility, many others feel constrained by a fixed learning path.
- The platform appears to support baseline usability, but not strong personalization or efficient shortcuts for users with different expertise levels.

Behavioural Analysis of the Users

The user behaviour suggested by these responses includes:

- Experienced users may feel slowed down when forced through content they already understand, reducing efficiency.
- Less flexible pathways can create a one-size-fits-all experience that does not align well with varied learning preferences.

- Mixed ratings indicate that users differ in the extent to which they feel the platform adapts to their skill level and pace.

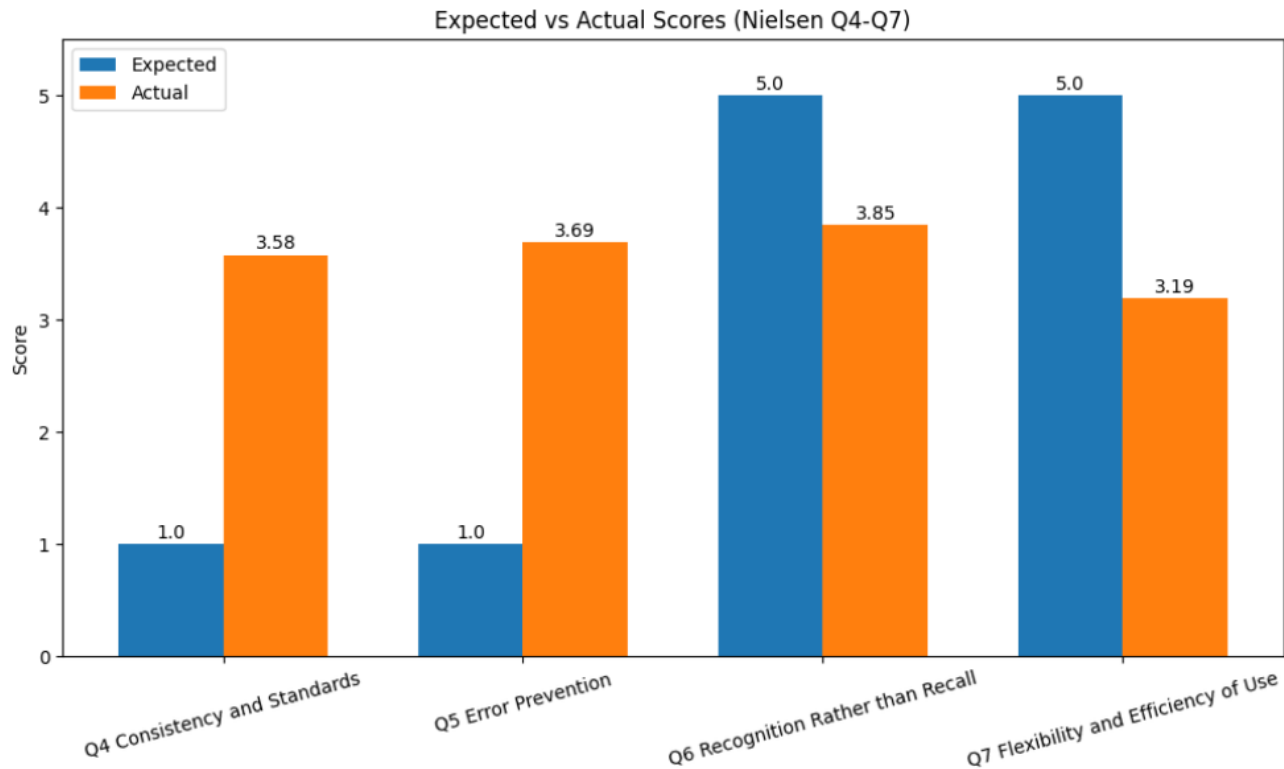


Figure 18: Expected vs actual comparison for Nielsen principles Q4–Q7

Comparative Inference

The expected-versus-actual comparison highlights that **Recognition Rather than Recall (Q6)** performs closest to the ideal usability condition, while **Consistency and Standards (Q4)** and **Error Prevention (Q5)** show the largest deviation from expectation. **Flexibility and Efficiency of Use (Q7)** occupies a middle position, suggesting partial fulfilment of the heuristic but limited support for adaptive or expert-friendly interaction.

Aesthetic and Minimalist Design

Q8. Have you ever felt overwhelmed by too many elements on the screen at once notifications, scores, timers, instructions competing for your attention?

Observations and Implications

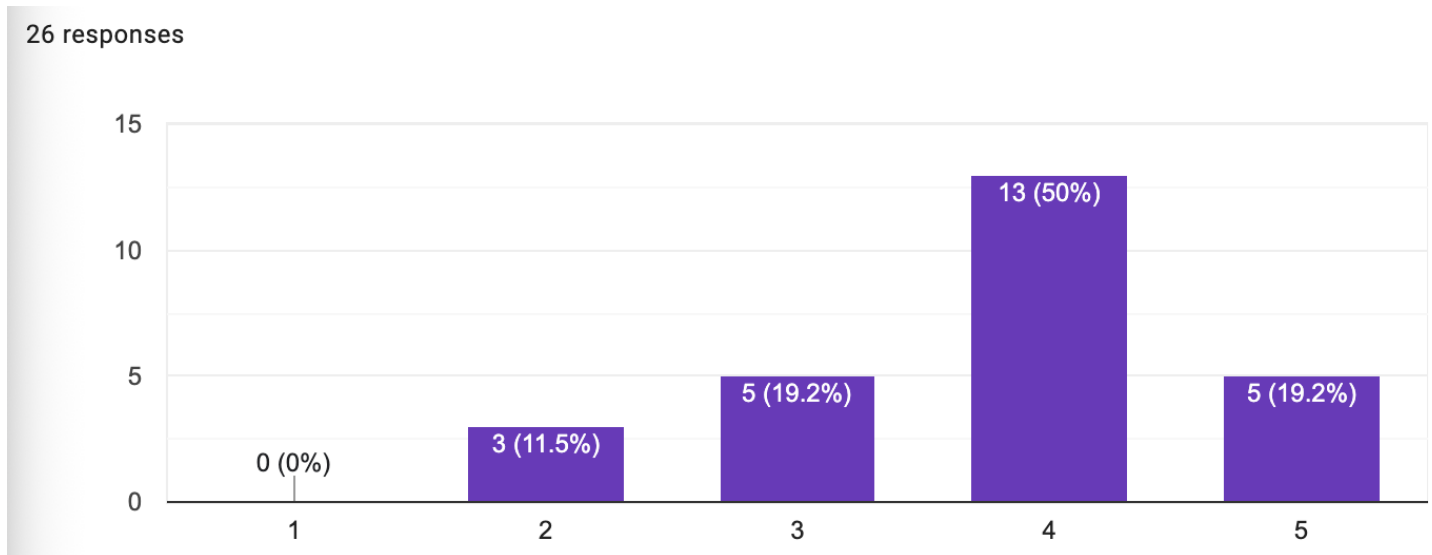


Figure 19: Participant responses

- A majority of users (50% rated 4) reported feeling overwhelmed by multiple on-screen elements, indicating a strong presence of visual clutter.
- A notable proportion of users (19.2% each for ratings 3 and 5) further reinforces that many users experience moderate to high levels of overload.
- Very few users rated low (11.5% at 2 and 0% at 1), suggesting that minimal clutter is rarely experienced.

Behavioural Analysis of the Users

The results reveal three distinct behavioural patterns among Coursera users:

- Users are likely experiencing cognitive overload, where excessive information competes for attention and reduces focus on the primary task.
- Many users may develop selective attention or ignore certain UI elements, potentially missing important information.
- The strong skew towards higher ratings indicates that users consistently perceive the interface as cluttered, which may lead to fatigue and reduced efficiency over time.

Help Users Recognize, Diagnose, and Recover from Errors

Q9. When something goes wrong (a wrong answer, a failed submission, a loading error), does the app explain what happened and how to fix it in plain language?

Observations and Implications

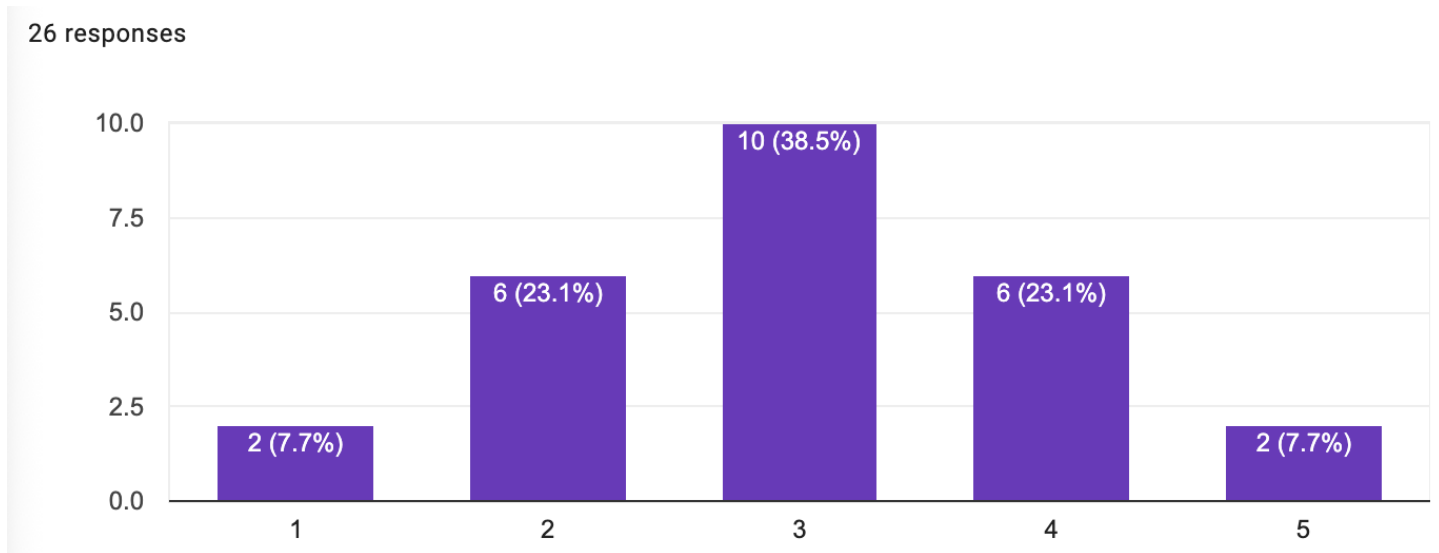


Figure 20: Participant responses

- The responses are centered around the mid-range, with 38.5% of users selecting 3, indicating a neutral or moderate experience with error explanations.
- A nearly equal proportion of users rated 2 (23.1%) and 4 (23.1%), suggesting inconsistency in how effectively errors are communicated.
- Only a small percentage of users rated the system at the extremes (1 and 5 at 7.7% each), indicating that very poor or very good experiences are less common.

Behavioural Analysis of the Users

The responses indicate the following behavioural patterns:

- Users tend to partially understand errors but lack complete clarity, leading to hesitation rather than confident recovery.
- Many users likely engage in trial-and-error behaviour due to insufficient or unclear guidance from the system.
- The variation in responses suggests that error messaging is inconsistent, causing different experiences across users or scenarios.

Help and Documentation

Q10. If you were confused about how a feature works, was it easy to find guidance within the app or did you have to search outside of it (YouTube, forums, asking someone else)?

Observations and Implications

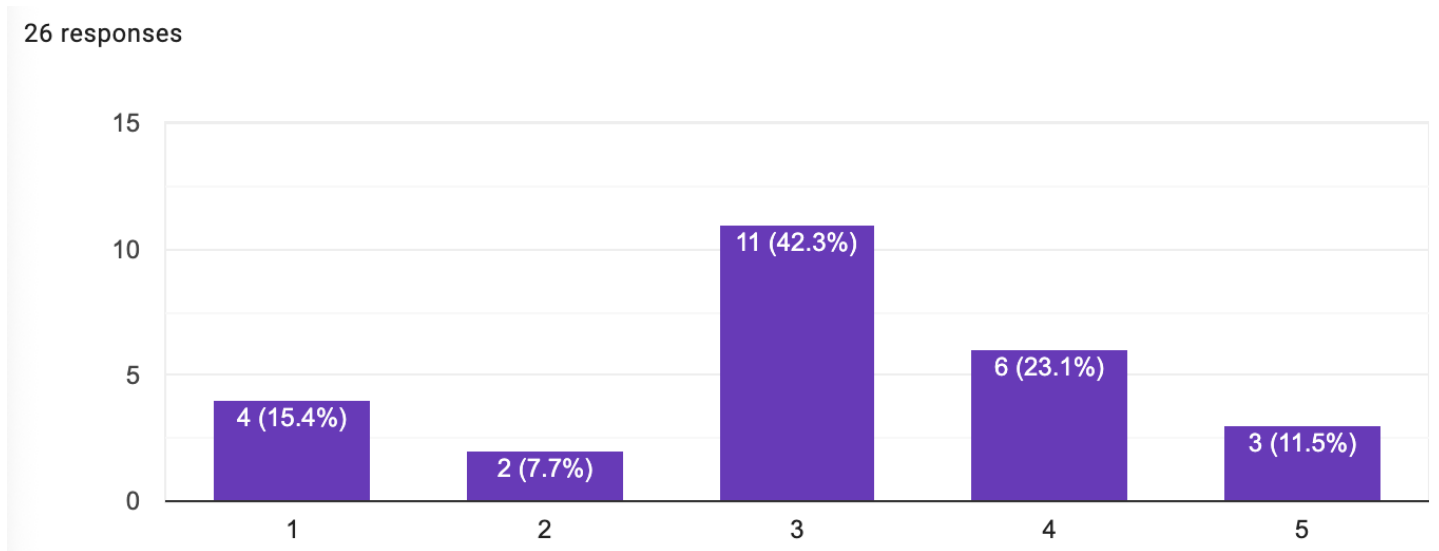


Figure 21: Participant responses

- The majority of responses are centered around rating 3 (42.3%), indicating a neutral experience in finding help within the app.
- A significant portion of users (23.1% rated 4) found it relatively easy to access guidance, but this is not dominant.
- Notably, 23.1% of users rated low (1 and 2 combined), suggesting that a considerable number of users struggle to find help and rely on external resources.

Behavioural Analysis of the Users

User behaviour suggests the following trends:

- Users often experience uncertainty when navigating features, leading to dependence on external platforms like YouTube or peers.
- The neutral majority suggests users eventually find help, but not efficiently, indicating poor discoverability rather than absence of help.
- A segment of users exhibits self-reliant or workaround behaviour, where they bypass in-app guidance and seek faster solutions outside the system.

Real world problems it runs into

Engagement & Motivation

Q1. When you feel stuck on a challenge in the app, what do you usually do – push through, skip it, or stop using the app for a while?

Observations and Implications

26 responses

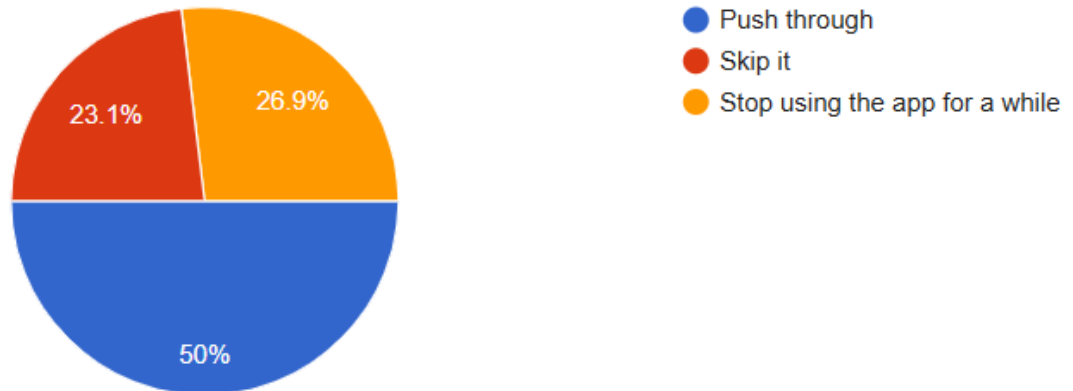


Figure 22: Participant responses

- Half the users push through challenges, which suggests reasonable frustration tolerance — but this also raises the question of whether the difficulty is appropriately calibrated. If users are forcing themselves through rather than enjoying the challenge, it indicates poor flow state design.
- About 27% of users stop using the app entirely when stuck. This points to a need for better scaffolding mechanisms like hints, checkpoints, or adaptive difficulty.
- Nearly 24 % users skip challenges. This breaks the intended progression graph, meaning users may miss prerequisite knowledge that might be required for other course.

Behavioural Analysis of the Users

The survey results reveal three distinct behavioural patterns among Coursera users when faced with challenges:

- The largest group, comprising exactly half of all respondents, demonstrates a persistence-driven mindset — these users push through difficulty, and have a higher tolerance for frustration, and a goal-oriented approach to learning. We can say that most of these users would have **high conscientiousness**.
- The second group represents the tendency of complete disengagement. When faced with a challenge, these users stop using the app entirely, indicating low frustration tolerance, or maybe a lack of confidence, or an absence of sufficient support within the platform to help them move forward. This behaviour is a strong predictor of permanent dropout, as temporary disengagement in digital learning environments frequently transitions into long-term abandonment.

- The third group exhibits avoidance behaviour by simply skipping the challenge altogether. While this keeps them on the platform, it reflects a superficial engagement pattern where the user prioritises progress over genuine understanding, ultimately leading to knowledge gaps that may hinder their learning outcomes over time.

Q2. Have you ever felt that earning points or badges stopped feeling meaningful? If so, when did that happen?

This was an open-ended question with the following highlights we obtained from the responses:

- **Lack of Real-World Value & Distraction**

The most prominent reason users cited for disengagement was the absence of any tangible, real-world utility attached to the points and badges they earned. Users recognised that Coursera's internal reward metrics carried no transferable weight outside the platform itself, like on LinkedIn or resumes. This reflects **instrumental value collapse** — when a reward loses its meaning because it cannot be exchanged for anything of genuine consequence. Users described the points as "just numbers". This is particularly significant on a platform like Coursera, where users are often working professionals or students with career-oriented motivations. When the rewards fail to align with those real-world ambitions, the disconnect becomes demotivating.

- **Ease of Acquisition & Trivial Content**

Another significant finding is that rewards lost meaning when they were perceived as too easy to obtain or were awarded for achievements that felt insignificant. When badges are handed out for minor milestones or after minimal effort, they fail to trigger the psychological satisfaction that comes from genuine accomplishment. This might be because intrinsic motivation is nurtured when individuals feel a sense of competence and mastery. If the tasks required to earn a badge are trivial, the badge itself signals low competence rather than real achievement, rendering it meaningless or even mildly insulting to more capable learners. Furthermore, when the course content itself is perceived as superficial, the entire reward structure built around it collapses in credibility.

- **Timeframe & Initial Perception**

Some users felt the rewards were meaningless from the very start, suggesting a segment of the user base that immediately identifies the surface-level engagement strategy with no real depth. These users likely entered the platform with high expectations and a clear sense of what genuine achievement looks like, and the reward system failed to meet that standard. On the other hand, users who experienced disillusionment after a few weeks or around two months describe an initial period where the rewards feel fresh and motivating, followed by a gradual erosion of that feeling as the repetitive and shallow nature of the reward structure becomes apparent.

Learning Effectiveness

Q3. Can you think of a moment when the game elements (scores, levels, rewards) distracted you from actually learning the material?

While game elements like scores, levels, and rewards are designed to motivate users to learn, the data reveals that for many respondents, these elements became the primary focus, effectively displacing genuine learning as the central goal.

- **Incentive Misalignment**

The pursuit of points, high scores, and level progression was the most commonly reported **source of distraction**. Users admitted to rushing through content, guessing answers rapidly, or seeking external help. This shows that a measure has become the target, and it isn't a good measure. Users

learned to optimise for the reward, creating a counterproductive loop where gamification incentivises behaviours that hollow out the learning experience.

- **Focus on Completion and Streaks**

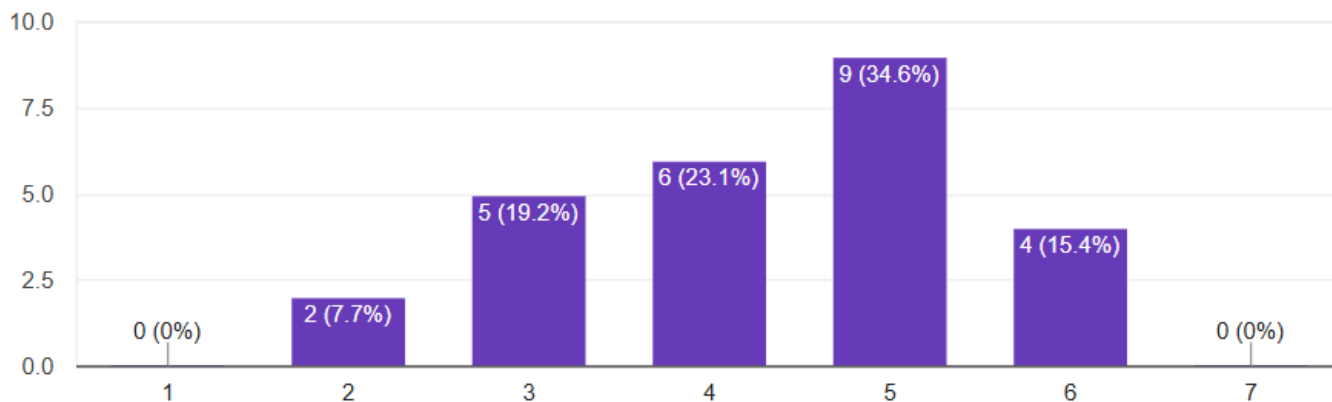
Several users reported rushing through material primarily to obtain a completion certificate before access expired or to maintain an unbroken streak, rather than out of any genuine desire to absorb the content. This shows how external pressures cause individuals to lose sight of the inherent value of the activity itself. Streaks in particular introduce a **time-based anxiety** that prioritises consistency of activity over quality of engagement, keeping users present on the platform. While this might be a good strategy for the app to keep their users, for a user it might lead to a negative result.

- **Negative Impact of Guessing**

When a user guesses correctly and is rewarded with points, the system reinforces a misconception — treating a lucky guess as equivalent to genuine knowledge, creating false confidence without real understanding. Similarly, timed quizzes and streak pressure introduce **cognitive load that diverts mental resources** away from thoughtful engagement and toward rapid, instinctive responses. The quiz mechanics end up being optimised for speed and engagement metrics rather than accurate assessment of comprehension.

Q4. How confident are you that what you practice in the app transfers to real-life situations or exams?

26 responses



The distribution skews noticeably toward the higher end of the scale, with the majority of responses clustered between ratings 4 and 6. This indicates that most users hold a moderate to moderately high level of confidence in the real-world transferability of their learning. But there is a lack of extremities, as in, no one had extreme certainty regarding the usefulness of Coursera in real life.

Access & Equity

Q5. How often you had to stop using the app mid-session because of a weak internet connection, a low battery, or a shared device?

Results: The mode is rating 2 (42.3%), but a substantial tail exists at ratings 4, 5, and 7. The mean is **3.00/7** ($n = 26$), showing that interruption burden is unevenly distributed across learners.

Table 1: Q5 response distribution (mid-session stoppages)

Rating	Count	Percentage
1	3	11.5%
2	11	42.3%
3	3	11.5%
4	3	11.5%
5	5	19.2%
7	1	3.8%
Total	26	100%

Graphical Observation

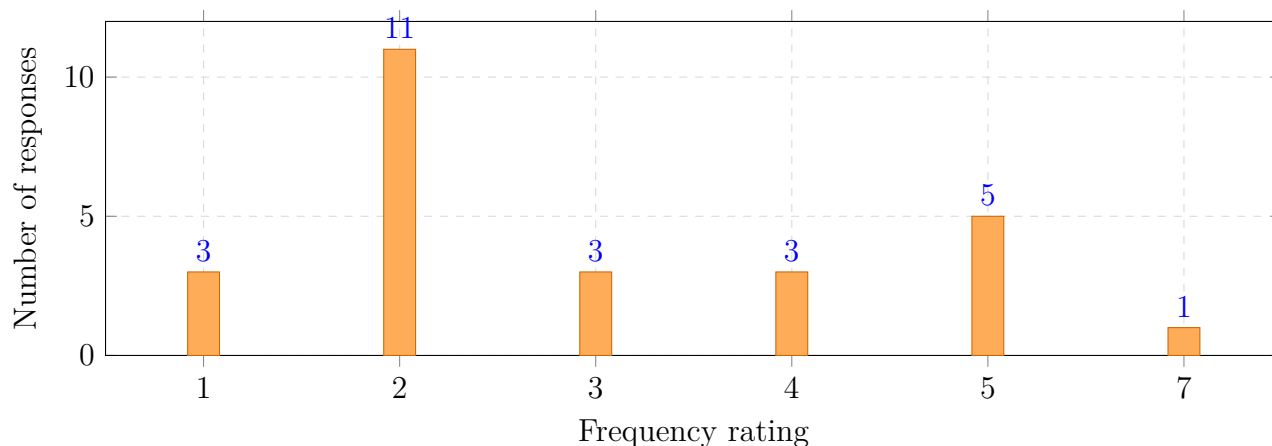


Figure 23: Q5 – stoppages from connectivity/device constraints (data-generated plot, $n=26$)

Inference - The core issue here is not average inconvenience, but **attention fragmentation inequality**. Even if many users report low-frequency interruptions, the high-interruption subgroup is repeatedly exposed to **task-switching costs** and **attention residue**, where part of working memory remains tied to the interrupted task after context shift. This weakens consolidation, especially in conceptually dense modules that require uninterrupted cognitive build-up. The pattern also maps to a real access-equity phenomenon: identical course design produces different learning trajectories because environmental reliability differs across users. In effect, infrastructural instability functions as an unmeasured extraneous load, reducing effective study time and creating avoidable variation in outcomes unrelated to ability or motivation.

Q6. Are there features or content in the app that feel designed for a different kind of learner than you in terms of language, cultural references, or learning pace?

Results: “Maybe” is the modal response (42.3%), followed by “Yes” (30.8%) and “No” (26.9%). This indicates ambiguity, not absence, of perceived learner-content mismatch.

Table 2: Q6 response distribution (learner-content mismatch)

Response	Count	Percentage
Yes	8	30.8%
Maybe	11	42.3%
No	7	26.9%
Total	26	100%

Graphical Observation

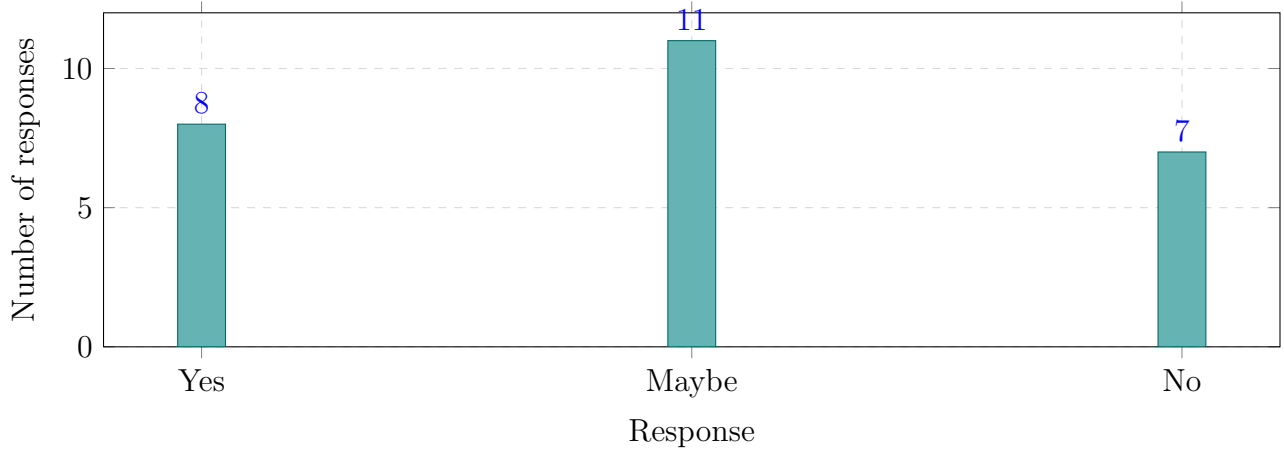


Figure 24: Q6 – perceived fit across language/culture/pacing dimensions (data-generated plot, n=26)

Inference - The dominant “Maybe” signal suggests **intermittent mismatch** rather than persistent exclusion, which is cognitively meaningful. When language register, examples, or pacing are only partially aligned with a learner’s prior schemas, the learner experiences increased **extraneous cognitive load** and reduced **processing fluency**. Even if comprehension eventually occurs, the additional decoding effort can suppress confidence, slow retrieval, and weaken transfer to new contexts. This pattern is also consistent with the **schema congruence principle**: content that maps onto familiar frames is encoded more efficiently and remembered more reliably. Hence, the data supports adaptive instructional framing (multi-level pacing, context-diverse examples, alternative explanation styles) to reduce avoidable cognitive friction for heterogeneous users.

Social & Competitive Dynamics

Q7. Have leaderboards or comparisons with other users ever made you feel discouraged rather than motivated? What did you do?

Results: Open-ended responses were coded into four sentiment groups to obtain interpretable structure.

Table 3: Q7 coded response summary (leaderboard/comparison effects)

Coded category	Count	Percentage
Not discouraged / no leaderboard exposure	14	53.8%
Motivated by comparison	4	15.4%
Discouraged or disengaged at least once	6	23.1%
Mixed/conditional self-regulation	2	7.7%
Total	26	100%

Graphical Observation

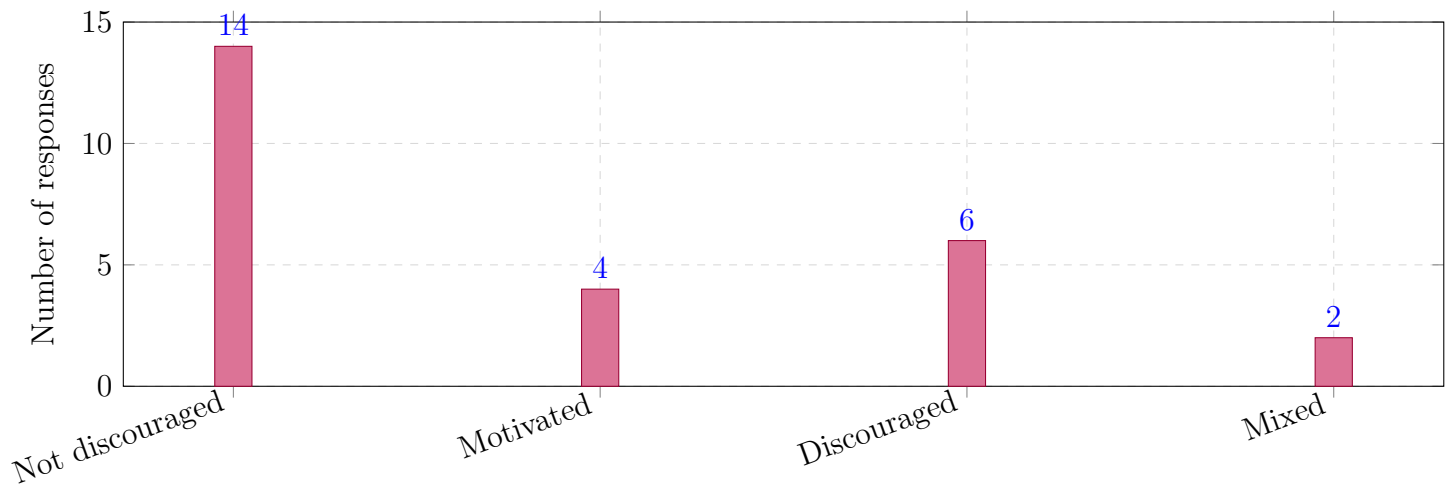


Figure 25: Q7 – coded sentiment toward social comparison (data-generated plot, n=26)

Summary of responses:

- **Limited direct exposure:** A notable set of users reported little or no leaderboard visibility on Coursera, so their responses reflect either neutral experience or transfer from other platforms.
- **Majority non-discouraged:** More than half did not report discouragement, indicating that social comparison is not universally harmful in this cohort.
- **Meaningful vulnerable subgroup:** Nearly one-fourth reported discouragement, temporary withdrawal, or rushing behavior when comparison pressure intensified.
- **Motivational heterogeneity:** A smaller group explicitly framed comparison as energizing and performance-enhancing.
- **Self-regulatory buffering:** Some participants described conscious reframing (focusing on personal learning) as a strategy to neutralize discouragement.

Inference - The distribution reflects classic **Social Comparison Theory** dynamics: upward comparison can either motivate effort or trigger inferiority and avoidance depending on perceived attainability. This is also compatible with **Self-Determination Theory**: competitive signals can support competence for some learners but undermine autonomy and intrinsic motivation for others when perceived as controlling. A third lens is the **Yerkes–Dodson relation**, where moderate arousal supports performance but excessive evaluative pressure degrades it, often producing speed-over-depth behavior. Hence, competitive elements should be personalized rather than globally imposed—for example by defaulting to self-referenced mastery trajectories and making public ranking opt-in—so the platform retains motivational benefits while minimizing discouragement and strategic shallow learning.

Q8. If you are using the app in a classroom or group setting, does the competitive structure cause tension between students?

Results: The modal response is “No” (46.2%), followed by “Maybe” (34.6%) and “Yes” (19.2%). This shows that overt conflict is not dominant, but latent or situational tension remains substantial.

Table 4: Q8 response distribution (competitive tension in group settings)

Response	Count	Percentage
Yes	5	19.2%
Maybe	9	34.6%
No	12	46.2%
Total	26	100%

Graphical Observation

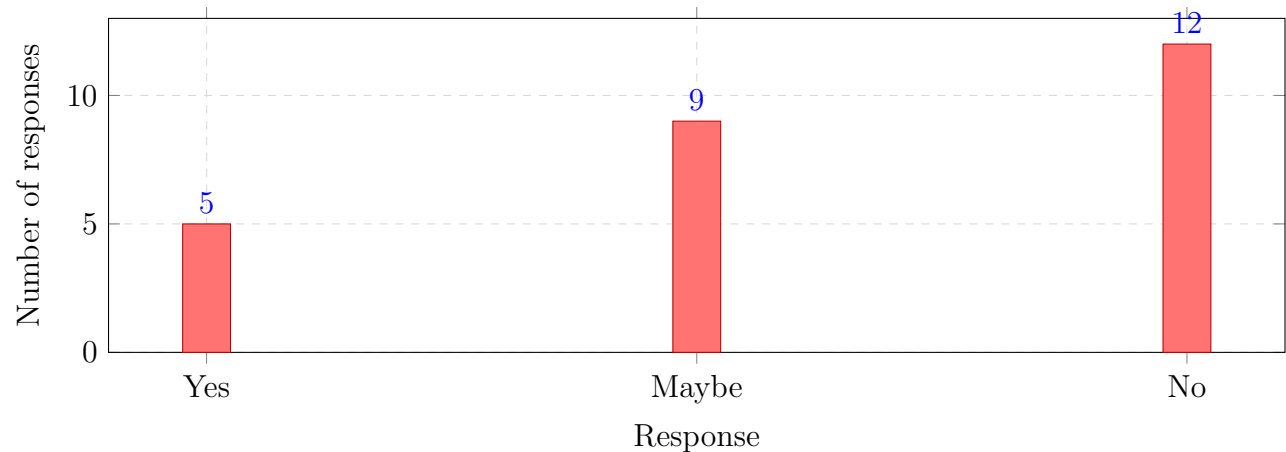


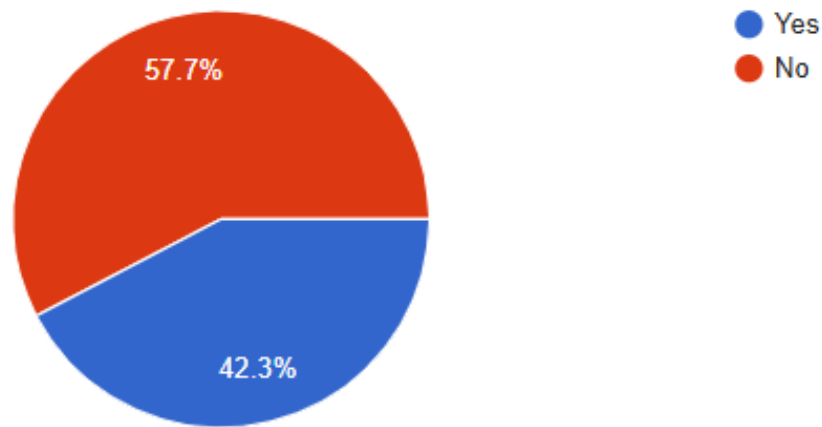
Figure 26: Q8 – perceived tension from competitive structure in classroom/group settings (data-generated plot, n=26)

Inference - The distribution indicates a **conditional social-friction effect**: although most users do not consistently experience tension, over half are either uncertain or explicitly report it in at least some contexts. This is consistent with **evaluation apprehension** and **social facilitation/inhibition** dynamics—performance pressure in visible comparison environments can improve output for confident learners but undermine participation quality for those who fear negative judgment. The high “Maybe” share also suggests that tension depends on contextual moderators such as group norms, instructor framing, and whether competition is perceived as fair and mastery-oriented. From a cognitive-social perspective, this pattern can be interpreted through **goal interdependence**: when peers are framed as rivals for rank, cooperative knowledge sharing declines and defensive behavior can increase. Therefore, classroom-facing competitive mechanics should be softened with team-based goals, private progress channels, and mastery-referenced feedback so motivation benefits are retained without escalating peer tension.

Trust & Long-Term Use

Q9. Have you ever questioned whether the app was tracking your progress accurately for example, marking something wrong that you believe was right, or not saving your work?

26 responses



42.3% (nearly half) of respondents reported concerns regarding the accuracy of progress tracking, including instances of perceived inconsistencies or failure to save progress.

Observations & Analysis

- **User-System Disconnect:** A substantial segment of the user base perceives a misalignment between their performance and the platform's evaluation. This undermines a user's sense of competence—a critical component of sustained engagement.
- **Increased Cognitive Load:** These incidents impose unnecessary extraneous cognitive load, e.g. more *recall rather than recognition*. Users are forced to divert mental effort from course material toward troubleshooting the platform, which diminishes overall learning efficiency.
- **Impact on Performance and Usage:** Consistent exposure to these system errors causes frustration and forces learners to not rely on the platform's metrics. They may experience decreased confidence in their own comprehension, potentially leading to lower motivation or increased abandonment rates.

Recommendations

- **Diagnostic Review:** The team can conduct an audit to determine if these reports stem from technical bugs or ambiguous UI design that misleads user input.
- **Enhanced Feedback Mechanisms:** The team can then implement more granular explanations for incorrect answers to help users reconcile their reasoning with the system's logic.
- **Reporting Channels:** They can provide a direct, low-friction mechanism for users to flag potential errors, transforming technical frustration into actionable insights for the development team.

Q10. What would make you stop using this app permanently, even if you found it useful at first?

Response Category	Count
Content Quality & Relevance	9
Monetization & Ads	7
Other/None	6
Professional/Academic Value	2
Technical Performance	2
Engagement Saturation	2

Summary of Responses

- **Content Relevance & Efficacy:** Users expressed concern over outdated material, poor quality content, lack of real-world applicability, and insufficient skill development. Some users also noted they would stop using the app if resources became available online for free.
- **Monetization:** Many users reported excessive charges, a paywall, the habit of asking money for every course, and the cessation of course scholarships to be a major cause if they were to abandon the app permanently.
- **Professional & Academic Value:** Abandonment is likely when certifications lose professional/-market standing or better options become available like finding a better app or an equally useful and comprehensive in-person learning environment.
- **Engagement Saturation:** Excessive promotion of paid content, unnecessary ads for upgrades and gamification elements (leaderboards, badges) causing interference often lose their efficacy over time, becoming perceived as "hollow" or distracting.
- **Technical and Performance Issues:** Stopping factors include the app being slow or laggy.

Inference Monetization strategies (ads, paywalls, upgrade prompts) function as "cognitive interrupters." In a learning context, maintaining a *flow state* is essential for comprehension. These interruptions increase cognitive load, diverting mental resources from the the learning material to managing unnecessary components. The reliance on gamification and extrinsic rewards creates a temporary engagement boost that masks a lack of deep, intrinsic value. When novelty fades, users perform a cost-benefit analysis and, finding the content lacks sufficient professional or practical depth, they churn. When content quality or technical performance fails, the perceived value collapses against free alternatives.

Net Promoter Score

Q11. How likely are you to recommend this app to a friend? (on a scale of 1 to 10 where 10 is most likely and 1 is least likely)

Total Responses = 26

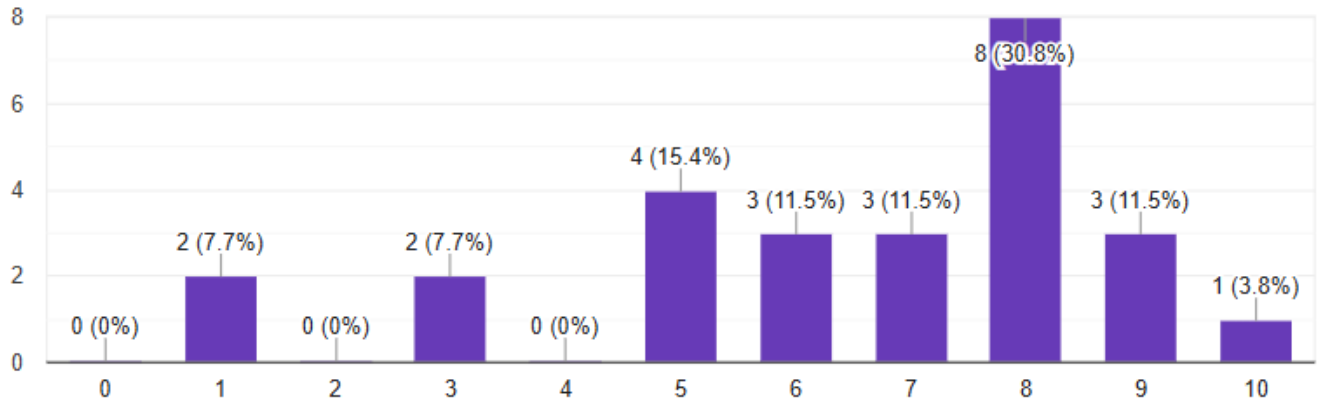
$$\text{Detractors [rated 0–6]} = 11 \quad \left(\frac{11}{26} \times 100 = 42.31\% \right)$$

$$\text{Passives [rated 7–8]} = 11 \quad \left(\frac{11}{26} \times 100 = 42.31\% \right)$$

$$\text{Promoters [rated 9–10]} = 4 \quad \left(\frac{4}{26} \times 100 = 15.38\% \right)$$

$$\begin{aligned} \text{Net Promoter Score (NPS)} &= \% \text{Promoters} - \% \text{Detractors} \\ &= 15.38 - 42.31 = -26.92 \end{aligned}$$

26 responses



Observations & Analysis

- **High Passive/Detractor Ratio:** The lack of enthusiastic promoters indicates that the app currently operates in a state of "functional utility" rather than building strong user advocacy.
- **Fragmented Sentiment:** The data displays a bimodal-like concentration where, despite a plurality of users rating the app an 8 (30.8%), a significant percentage (42.3%) fall into the "Detractor" category.
- **Low Advocacy Readiness:** An NPS of -26.92 suggests that the current user base is more likely to be neutral or dissatisfied than to engage in positive word-of-mouth indicating the app meets basic requirements but fails to delight the user.

Recommendations

- **Convert the Passives:** Try to engage the 42.3% of "Passives" (scores 7-8) with qualitative follow-ups and identifying the specific problems they face/ask them areas of improvement for increasing promotion.
- **Stabilize the Detractors:** Addressing the primary "Detractor" triggers—monetization friction, lack of profession value, content relevance—is essential before attempting to drive new growth.
- **Real-World Applicability:** Integrate "Real-World Use Case" modals for abstract concepts, providing links to professional applications or case studies to bridge the gap between app practice and real-life utility.
- **Revision through app:** Introduce algorithmic re-introduction of concepts (2, 7, and 30-day intervals) to combat the forgetting curve.
- **Hint-Hierarchy Feedback:** Replace binary "Correct/Incorrect" outcomes with staged hints, fostering active problem solving before revealing the answer.
- **Focus Mode:** Introduce a toggleable UI that hides non-essential elements (leaderboards, timers, social notifications) during study sessions to minimize extra cognitive load.

Conclusion

Key Takeaways

Memory and Learning Science

Coursera performs reasonably well at the initial encoding stage: a majority of users (57.7%) found new material clearly explained, and the platform’s use of both verbal and visual content aligns with Paivio’s Dual-Encoding Hypothesis. Gamification elements such as points, streaks, and badges achieved a moderate mean engagement score of 3.73/5, consistent with Processing Efficiency Theory. However, the platform exhibits critical weaknesses in consolidation and long-term retention. Spaced repetition is largely absent—80.8% of users reported only *occasional* revisitation of prior material, with no systematic review schedule in place. Review sessions are perceived as passive (mean 3.31/5), relying on rewatching lectures rather than the generative retrieval that the Testing Effect demands. The most alarming finding is a mean confidence score of only 2.96/5 for two-week retention, the lowest across all memory-related items, directly reflecting Ebbinghaus’ forgetting curve in action. Metacognitive progress tracking is present but shallow: while all users reported some visibility of progress, half found it insufficiently granular to support accurate self-assessment (Flavell, 1979).

Nielsen’s Usability Heuristics

Among the ten Nielsen heuristics examined, the platform performs best on **Recognition Rather than Recall** (mean 3.85/5), suggesting that structured navigation and progress cues allow learners to resume sessions with low cognitive overhead. **User Control and Freedom** is also largely adequate, with 15 out of 26 users rating undo and back-navigation positively.

In contrast, the platform shows its weakest adherence to **Aesthetic and Minimalist Design**—half of all respondents rated visual overload at the maximum level, pointing to interface clutter that competes for attentional resources and risks inducing cognitive overload. **Consistency and Standards** (mean 3.58/5, negatively framed) and **Error Prevention** (mean 3.69/5, negatively framed) also reveal meaningful gaps: users encounter inconsistent interface behaviour across sections and are insufficiently protected from accidental actions. **Match Between System and the Real World** is particularly polarised (42% negative ratings), indicating that gamified terminology alienates a significant portion of the user base by misaligning with their existing mental models. Help and documentation, while generally reachable, suffers from poor discoverability, pushing users toward external resources such as YouTube and peer discussion.

Real-World Problems

The real-world analysis reveals that Coursera’s user experience is shaped as much by systemic design tensions as by individual preferences. On engagement, half of users push through difficulties, but roughly a quarter disengage entirely—a pattern consistent with the absence of adaptive scaffolding and well-calibrated flow states. Gamification rewards suffer from instrumental value collapse: users, especially those with professional or career motivations, quickly recognise that points and badges carry no transferable external value, leading to erosion of extrinsic motivation within weeks. Critically, game mechanics such as streaks and timed quizzes introduce time-based anxiety that incentivises speed over comprehension, reinforcing shallow learning rather than deep understanding.

Access and equity emerge as structurally important variables. Connectivity and device constraints unevenly fragment learning sessions, and over 73% of users reported at least partial mismatch between the platform’s language, cultural references, or pacing and their own learner identity. Social comparison via leaderboards motivated some users while discouraging others, consistent with Social Comparison Theory and Self-Determination Theory: competitive mechanics require personalisation rather than global imposition.

Finally, 42.3% of users expressed doubts about the accuracy of progress tracking, generating extraneous cognitive load and eroding trust in the platform’s evaluative reliability. The survey concluded with a Net Promoter Score of **-26.92**, reflecting a user base in a state of functional tolerance rather than genuine advocacy, with monetisation friction and limited real-world applicability identified as the primary churn triggers.

Learning from the Project

This project offered our group a hands-on understanding of how cognitive science principles apply—and how readily they can be violated—in everyday digital products that millions of learners use.

The gap between intent and implementation is large. Coursera is widely regarded as a high-quality platform built in collaboration with world-class universities, and yet our analysis revealed systematic misalignments with well-established principles of learning science and usability. This taught us that good content and good cognitive design are not the same thing, and that even reputable platforms leave substantial room for improvement.

User behaviour is heterogeneous and context-dependent. One of the most repeated lessons across sections was that no single design decision affects all users the same way. Gamification motivates some and alienates others; leaderboards energise some and discourage others; spaced review helps some but goes unnoticed by those who are not intrinsically motivated. Effective platform design must therefore embrace adaptive personalisation rather than uniform experience.

Interdisciplinary thinking is necessary. No single framework—not memory science, not heuristic evaluation, not motivation theory—was sufficient on its own to explain Coursera’s strengths and weaknesses. The richest insights emerged when we triangulated across frameworks: for instance, recognising that low retention confidence (memory science), passive review formats (elaborative retrieval theory), and lack of spaced repetition (Spacing Effect) were all facets of the same underlying problem. This kind of integrative analysis is, we believe, the distinctive value that cognitive science brings to product evaluation.

In sum, this project demonstrated that the principles of cognitive science are not merely academic—they are actionable design criteria whose presence or absence has measurable consequences for how effectively people learn. Platforms like Coursera that take these principles seriously from the ground up, rather than applying them as surface-level features, stand to produce genuinely better learning outcomes for their users.